

Ronish Bhatt

Software engineer · Toronto, Canada · open to remote

ronishbhatt.com github.com/sunmoonron [nostr: ronish@ronishbhatt.com](mailto:ronish@ronishbhatt.com)

SUMMARY

Honours BSc in Computer Science (software engineering co-op, University of Toronto, 2025). I build full-stack products and run the infrastructure under them, and I find bugs in other people's systems. This year: a website whose pages are signed Nostr events served from my own relay, a proof-of-work community layer that needs no accounts, a Bitcoin timechain analytics tool with a from-scratch 3D renderer, and disclosures to Anthropic and OpenAI.

SKILLS

languages	TypeScript, JavaScript, Python, Java, SQL, Nix
web	Node, Preact and React, NestJS, Django, Canvas 2D, Web Workers, static and edge hosting behind Cloudflare
infrastructure	NixOS, Linux, nginx, Docker, systemd, Cloudflare Tunnel, AWS, Azure DevOps; declarative, reproducible hosts
bitcoin, nostr	Bitcoin Core and strfry relay operations, relay write-policy plugins, NIPs 01, 05, 09, 13, 17, 44, 59, 96 and 98 by hand, difficulty-epoch timechain math
security, ml	responsible disclosure, API and protocol testing, Charles Proxy, threat modelling; PyTorch, transformers, mechanistic interpretability, pandas, scikit-learn

SELECTED WORK, 2026

ronishbhatt.com, a website made of signed events 2026

- A 22-line HTML shell; layout, stylesheet and every content block are signed Nostr events fetched from a personal strfry relay with public relays as backups, verified and assembled in the browser, with subresource and import-map integrity end to end. Content veiled with NIP-44 under a key that is the favicon itself (an 8×8 mosaic encoding 32 bytes).
- NIP-17 gift-wrapped chat and small experiments gated by NIP-13 proof of work instead of accounts, enforced by tiered relay write policy. Runs on a hardened NixOS host with Bitcoin Core and the relay behind Cloudflare Tunnel, operated through a zero-SSH dashboard.

NOSTR ORBIT, a Nostr history on the timechain 2026

- Maps any profile's events to Bitcoin block heights using the real difficulty schedule (about five blocks of error instead of tens of thousands from a naive average), with halving-era rings and NIP-03 attestation counts.
- A voxel chain where every Bitcoin block is a cube (12×12 a day, 14 days an epoch, a coil turn a year), mempool-style epoch blocks whose contents are the notes as a treemap coloured by relay-reported engagement, and a scrubbable replay. Software 3D on a 2D canvas, no libraries.

Doorway and Toronto Skating 2025 to 2026

- Live rink schedules for a city with a Nostr community layer (chat, comments, votes) that needs no sign-ups; the proof-of-work relay policy behind it is now its own open-source project, github.com/sunmoonron/doorway.

Plant a Thought, and interpretability research 2026

- A 7 M-parameter language model and a tiny inference engine running entirely in the browser, taught a memory during the session. Independent papers on weight dictionaries and superweight circuit patching in small transformers; a public context-dilution demo on GPT-2 with real logits in the browser.

SECURITY RESEARCH (INDEPENDENT)

Responsible disclosure 2018 to present

- Reported a distillation and unmetered-compute issue in an Anthropic API endpoint through their security program; triaged as informative.
- Reported sandbox infrastructure and credential exposure in OpenAI systems through their security program; triaged as informative.
- Disclosed a double-spend-class race condition at a major crypto platform (under NDA), a promotional-credit accrual loophole in a large e-commerce mini-app, and a pack-opening exploit in a Web3 card game; vendors patched.

OPEN SOURCE

- **pandas** (2023): implemented `escape="latex-math"` for LaTeX table rendering so math mode survives while unsafe characters are escaped (issue #50040), with tests across edge cases; co-authored the test plan for `merge(..., combine=...)` (#16070).

EXPERIENCE

Financial Technology Risk Analyst (co-op)

May to Aug 2021

Office of the Superintendent of Financial Institutions, Toronto

- Built the Power BI and Excel dashboard for 320+ cyber incidents across regulated institutions, cutting redundant categories by a fifth; an NLP pipeline (sentiment and topic modelling) surfaced emerging risks and led to three new controls.
- Automated MISP deployment on Ubuntu, taking setup from three days to three hours.

Web Developer (co-op)

Jan to Apr 2022

Media Matters, Toronto

- Shipped six responsive React pages that lifted average session duration by about 15 percent; automated the HubSpot hand-off, cutting lead-routing time by a quarter; scraped 1,000+ dealership sites to feed a market-analysis dashboard.

EDUCATION

Honours Bachelor of Science, Computer Science

2019 to 2025

University of Toronto Scarborough · Specialist in Software Engineering, co-op · University of Toronto Scholar Award